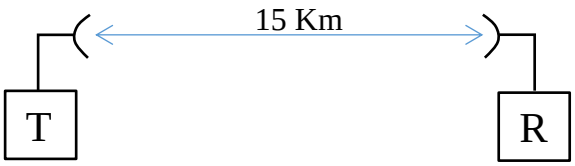
Exercise 2

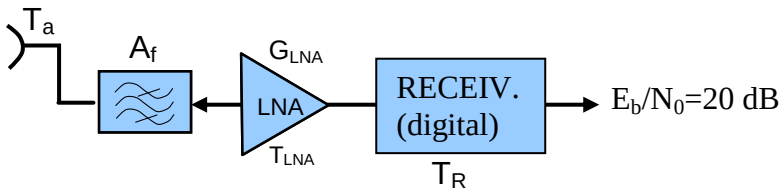


The figure shows a terrestrial link operating at 12 GHz with bandwidth B=10 MHz. The two stations are located 15 Km away and the antennas are identical with the following features:

Efficiency η=0.8. Directivity function $f(\theta)$=1 for $0< \theta <20°$, =0 elsewhere.

1) Evaluate the gain G of the antennas
2) It is known the system noise temperature Tsys=1000 °K of the station R. What it the power to be transmitted from R in order to get the system SNR equal to 30 dB? Assume the antennas pointed for the maximum gain.

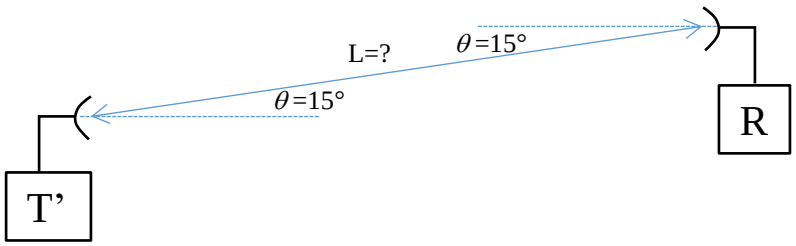
Consider the following scheme for the receiving station R:



T_{LNA}=150°K
T_a=500°K
G_{LNA}=10 dB
A_f=1 dB
T_R=? °K

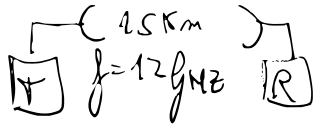
3) Evaluate the equivalent noise temperature of the receiver (T_R) compatible with the imposed system SNR .
4) Assuming E_b/N_0=20 dB at the receiver output, what is the maximum data rate R?

Consider now another transmitting station located as in the following figure, transmitting a power P=10W at the same frequency of T (12 GHz). The antenna of this station is characterized by G=10 dB and $g(\theta)=\cos(5.7\cdot\theta)$. The signal received by R represents in this case an interference and should be treated as noise which is added to the thermal noise of the receiving station.



5) Evaluate the minimum distance L so that the overall system SNR of R is not lower than 20 dB.

EXERCISE (23/02/2015)



$BW = 10\,MHz$

ANTENNAS are EQUAL

$\eta = 0.8$; $f(\vartheta,\varphi) = \begin{cases} 1 & \vartheta \geq 0 \\ 0 & \vartheta > 0 \end{cases}$

(1) G? (2) $T_{SYS} = 1000\,K$ for RECEIVER; $SNR = 30\,dB \rightarrow P_{TX} = ?$

(1) $S(r,\vartheta,\varphi) = \dfrac{P_{RAD}\, D_M \cdot f(\vartheta,\varphi)}{4\pi r^2}$ $\qquad P_{RAD} = \oint\!\!\oint S_{RAD}\, dS$

$= \dfrac{P_{RAD}\, D_M}{4\pi r^2} \displaystyle\int_0^{\pi}\int_0^{2\pi} f(\vartheta,\varphi)\, r^2 \sin\vartheta \, d\varphi d\vartheta$

$$\Rightarrow D_M = \frac{4\pi}{\displaystyle\int_0^{\pi}\int_0^{2\pi} f(\vartheta,\varphi)\sin\vartheta\, d\varphi d\vartheta} = \frac{4\pi}{2\pi \displaystyle\int_0^{20^\circ} \sin\vartheta\, d\vartheta} = \frac{4\pi}{2\pi\left[-\cos\right]_0^{20^\circ}} = \frac{4\pi}{2\pi(1-\cos 20^\circ)}$$

$= 33 = 15.2\,dBi$

$G = \eta \cdot D = 14.2\,dBi$ $\qquad k = 1.38 \cdot 10^{-23}\ \frac{W}{Hz\,K}$

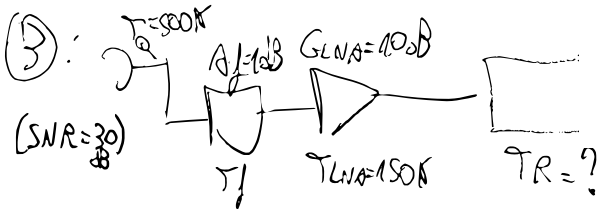
(2) $P_{NOISE} = k\, T_{SYS}\, B = 0.138\,pW \Rightarrow P_{RX} = P_{NOISE_{dB}} + SNR_{dB} = -68.6\,dBm$

$(= -98.6\,dBm)$

$P_{RX}\,[dBm] = P_{TX}\,[dBm] + G_{TX}\,[dBi] - FSL\,[dB] + G_{RX}\,[dBi]$ $\qquad FSL = -20\log\frac{\lambda}{4\pi d} = 137.5\,dB$

$P_{TX} = P_{RX} - G_{TX} - G_{RX} + FSL = 40.5\,dBm = 10.5\,dBW = 11.2\,W$

(3):



$$T_{eq} = T_a + T_f + \frac{T_{LNA}}{G_f} + \frac{T_R}{G_f G_{LNA}} = 1000\,K$$

$$SNR = \frac{P_S}{P_N} = \frac{\frac{E_S}{T_S}}{P_N} = \frac{\frac{E_b}{T_b}}{N_0 \cdot B} = \frac{E_b \cdot R_{BS}}{N_0 B} \Rightarrow \frac{R_{BS}}{B} = \frac{SNR}{E_b/N_0} = 10$$

$$T_R = G_f G_{LNA} \cdot \left(T_{eq} - T_a - T_f - \frac{T_{LNA}}{G_f} \right)$$

$G_f = 0.9772$ $\quad T_f = T_0(NF_f - 1) = T_0(A_f - 1) = 75.1\,K$

$G_{LNA} = 10$

$T_R < 2652\,K$ $\quad$ 1

(4) if $E_b/N_0 = 20$ dB, what is the DATA RATE?

$$SNR = \frac{E_b}{N_0} \frac{R_{BS}}{B} \Rightarrow R_{BS} = B \cdot \frac{SNR}{E_b/N_0} = 100\,Mbps$$

(5): Another TX is INTERFERING:



T has $G = 10$ dB, $f(\vartheta,\varphi) = \cos(5.7\vartheta)$

⇒ What is the minimum L to have $SNR_{dB}^{(R)} = 20$ dB

$P_{INT} = P_{TX} - FSL + G + 10\log\cos(5.7 \cdot 0.26\,rad) + G$ $\quad$ $100 = \frac{P_{RX}}{P_N + P_{INT}}$ $\quad$ ($P_{RX} = -68.6$ dBm, $P_N = -98.6$ dBm)

$P_{INT} = 1.24\,nW\ (= -89.0\,dBm) \Rightarrow FSL = P_{TX} + G - 11.1\,dB + G_R - P_{INT} = 142.6\,dB \Rightarrow d \geq 25.3\,Km$